

Case Studies in Engineering

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1 Preface

This document contains all my internal monologues, thoughts and learnings while going through various problems in engineering. Not sure who else is gonna read this but I hope you're not bored out of your mind reading noob stuff.

2 ML Experiments

2.1 Benchmarking OCR model on shipment packing list data

In logistics domain, shipment packing lists are essential documents that detail the contents of a shipment. Currently, the information from the packing list is manually entered into the system which is time-consuming. To automate this, we experimented with various OCR models to extract text from scanned packing list images.

2.1.1 Problems in Packing List text extraction

Inconsistent Data Format Packing lists vary significantly in format, font styles, and layouts across different vendors. Some issues we observed in collected data is:

- Different layouts, fonts and structures
- Tables spanning multiple pages
- Handwritten notes and stamps overlapping printed text
- Low-quality scans with noise and skew
- Varied languages and character sets
- Complex tables with merged cells and inconsistent alignments

2.2 Solution #1: Text extraction through LLM-models

We experimented with LLM-based models like GPT-4 and Claude to extract structured data from packing list images. The workflow involved:

1. Upload the scanned packing list image + prompt.
2. Use LLM to extract raw text from the image.
3. Use LLM to parse the extracted text into structured JSON format.
4. Post-process the JSON to ensure data consistency.

Assumptions about this approach This is not a traditional OCR approach. We are leveraging the LLM's ability to understand context and structure in text. So we have to assume

- This is the most viable approach given the business needs of quick turnaround and minimal manual intervention.
- The cost of using LLMs is justified by the time saved in manual data entry.
- The LLMs can recognize and interpret various formats and layouts at an acceptable accuracy level.
- This approach is only to validate the idea before investing in building a custom OCR solution.